



[Home](#) | [About Us](#) | [Success Stories](#) | [Farmers Association](#) | [Farmers' Innovation](#) | [Publications](#) | [Contact](#)

Post Harvest Diseases:: Vegetables:: Tomato

Early Blight: *Alternaria solani*

Symptom:

- The fruit become infected-through the calyx or stem attachment, either in the green or ripe stage. Concentric ring present on the fruit surface.
- Appear leathery and may be covered by a velvety mass of black spores.
- Infected fruit frequently drop, and losses of 50% of the immature fruit may occur.



Early blight



Black spores



Infected tomato

Identification of pathogen:

Fungus:

- Mycelium- septate, branched, light brown, become darker with age
- Conidiophores- 50 to 90µm in length and dark coloured
- Conidia-beaked, muriform, dark coloured and borne singly

Favourable condition:

- Maximum dispersal of conidia- 9am and 12 noon
- plants suffering from water stress- susceptible

Spread and survival:

- Spread by wind and rain splashes
- Under dry conditions- survives in infected plant debris in the soil for 3 yrs
- Seed borne

Management:

- Use disease free seeds for sowing
- Field sanitation
- Crop rotation with non solanaceous crop
- Optimum irrigation- to avoid stress condition
- Seed treatment- thiram 2g/kg
- 3 sprays with difolatan 0.2 % at fortnightly interval

Content validator:

Dr. M. Deivamani, Assistant Professor, Horticulture Research Station, Yercaud-636602.

Source of Images:

http://wiki.bugwood.org/Alternaria_solani_%28Early_blight_of_tomato%29

<http://www.apsnet.org/edcenter/intropp/lessons/fungi/ascomycetes/Pages/PotatoTomato.aspx>